

RBF-FD Solver: Least Squares Method for Operator Reproduction

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Updated: 9/16/25

Contents

1	Problem Setting	2
2	RBF-FD Method	3
2.1	Local Approximation setup	3
2.2	Polynomial Augmentation	3
2.3	Consistency Error Analysis	4
3	Weight Constraint RBF-FD Setup	5
4	Consistency Conditions for Finite Spaces and Interpolation Results	6
5	Consistency Results for Function Approximations	7

1 Problem Setting

Following the standard RBF-FD methods, let $u \in \mathcal{F}$ be a solution for a PDE in form of:

$$-Lu = -\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega, \quad (1)$$

Let $1 < p < \infty$, $A \in W^{1,\infty}(\bar{\Omega}; \mathbb{R}^{d \times d}) : \bar{\Omega} \rightarrow \mathbb{R}^{d \times d}$ is an symmetric positive definite anisotropic tensor, where

$$\eta |\xi|^2 \leq \mathbf{A}(\mathbf{x}) \xi \cdot \xi \leq \mathbf{H} |\xi|^2, \quad \mathbf{0} < \eta \leq \mathbf{H} \quad (2)$$

Assume $\Omega \subset \mathbb{R}^d$ and $f \in W^{k,p}(\Omega)$ is the forcing term acting in the domain Ω . The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (3)$$

$$n \cdot A \nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (4)$$

For simplicity, we assume Ω be a bounded $C^{1,1}(\partial\Omega)$ domain, with $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. Furthermore, let $g_D \in W^{k-1/p,p}(\partial\Omega)$, $g_N \in W^{k-1-1/p,p}(\partial\Omega)$. By *Agmon-Douglis-Nirenberg* Theorem [2], we have the following estimate for u :

$$\|u\|_{W^{k,p}(\Omega)} \leq C \left(\|Lu\|_{W^{k-2,p}(\Omega)} + \|g_D\|_{W^{k-1/p,p}(\partial\Omega)} + \|g_N\|_{W^{k-1-1/p,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)} \right). \quad (5)$$

In the case that we work on a Hilbert space, we then have for $p = 2$ ($W^{k,2} := H^k$) the following estimate for u :

$$\|u\|_{H^k(\Omega)} \leq C \left(\|Lu\|_{H^{k-2}(\Omega)} + \|g_D\|_{H^{k-\frac{1}{2}}(\partial\Omega)} + \|g_N\|_{H^{k-\frac{3}{2}}(\partial\Omega)} + \|u\|_{L^2(\Omega)} \right). \quad (6)$$

If we want to study cases where $(Lu)(\mathbf{x}) \in L^\infty(\Omega)$. we can formulate the setup by having $f \in W^{s,p}(\Omega)$ and, by Morrey Embedding,

$$W^{s,p} \hookrightarrow C^{0,\alpha}, \quad \text{with } \alpha = s - \frac{d}{p} > 0 \quad (7)$$

In other words, $s > \frac{d}{p}$. Suppose $p = 2$, then $s > \frac{d}{2}$. Given an operator $L : H^{s+2}(\Omega) \rightarrow H^s(\Omega)$, we define an error functional of the form:

$$\xi_{L,x} : H^{s+2}(\Omega) \rightarrow \mathbb{R}, \quad \xi_{L,x} \in (H^{s+2})^*(\Omega) \quad (8)$$

Then by Riesz Representation Theorem, there exists a $v_{L,x} \in H^{s+2}(\Omega)$ such that $\xi_{L,x}(u) = \langle u, v_{L,x} \rangle_{H^{s+2}(\Omega)}$, and $\|\xi_{L,x}\|_{(H^{s+2}(\Omega))^*} = \|v_{L,x}\|_{H^{s+2}(\Omega)}$

2 RBF-FD Method

2.1 Local Approximation setup

To obtain an approximation of the operator L at point $\mathbf{z}$, we can approximate $\hat{u} \approx u$ on a neighborhood $\Omega \subseteq \mathbb{R}^d$ of $\mathbf{z}$ such that we approximate $Lu \approx L\hat{u}$. We define the basis functions in the form $\phi_i(\mathbf{x}) : \Omega \rightarrow \mathbb{R}$ for $i = 1, \dots, m$. Define the vector space $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega$ be a set of stencil nodes for a point $\mathbf{z}$ in Ω . For the classical approximation scheme, let $m = n$. We can now define $\Phi \in \mathbb{R}^{n \times m}$, where $\Phi_{ij} = \phi_j(\mathbf{x}_i)$. The role of these basis functions will be to approximate the operator L . The approximation of u takes the form of:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n \lambda_i \phi_i(\mathbf{x}) \quad (9)$$

This will lead to the form $\Phi \lambda = \hat{\mathbf{u}}$ for $\hat{\mathbf{u}} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. We can simplify for λ to obtain:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n (\Phi^{-1} \mathbf{u})_i \phi_i(\mathbf{x}) = \phi(\mathbf{x})^T \Phi^{-1} \mathbf{u} \quad (10)$$

The approximation of Lu at $\mathbf{z}$ is in the form of:

$$Lu(\mathbf{z}) \approx \mathbf{L} \hat{\mathbf{u}}(\mathbf{z}) = \mathbf{L} \phi^T(\mathbf{z}) \lambda = \mathbf{L} \phi^T(\mathbf{z}) \Phi^{-1} \mathbf{u} := \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) \mathbf{u} \quad (11)$$

Thus, approximation the operator turns into local solver for the weights w_i from the form:

$$\Phi^T \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) = \mathbf{L} \phi(\mathbf{z}) \quad (12)$$

Through the method of undefined coefficients on the basis $\{\phi_i\}$ we get the same resulting set of weights. Finally, by defining the basis for V_Y as radial basis functions, with $\phi_i(\mathbf{x}) := \phi(\|\mathbf{x} - \mathbf{y}_i\|)$ and centers $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega$. Thus we defined the Radial Basis Function generated Finite Difference method (RBF-FD).

2.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

$$\hat{u}(\mathbf{x}) = \sum_{j=1}^n \lambda_j \phi_j(\mathbf{x}) + \sum_{l=1}^l \beta_l \mathbf{p}_l(\mathbf{x}) \quad (13)$$

$$\sum_{i=1}^n \lambda_i p_j(\mathbf{x}_i) = 0, \forall j = 1, \dots, l \quad (14)$$

We can generate a Gram matrix equation of the form:

$$\begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \Phi & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \lambda \\ \alpha \end{bmatrix} \quad (15)$$

Hence, we can approximate the operator to take the form:

$$(L\hat{u})(\mathbf{x}) = \begin{bmatrix} (\mathbf{L}\phi)(\mathbf{x}) \\ (\mathbf{L}\mathbf{p})(\mathbf{x}) \end{bmatrix}^T \begin{bmatrix} \Phi & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} := \begin{bmatrix} \mathbf{w}(\mathbf{x}) \\ \nu(\mathbf{x}) \end{bmatrix}^T \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \mathbf{w}^T(\mathbf{x}) \hat{\mathbf{u}} \quad (16)$$

Thus, we get the local augmented operator reproduction scheme:

$$\begin{bmatrix} (L\phi)(\mathbf{x}) \\ (Lp)(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \Phi^T & P \\ P^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w}(\mathbf{x}) \\ \nu(\mathbf{x}) \end{bmatrix} \quad (17)$$

The polynomial constraint setup can be viewed as a Lagrange multiplier to a minimization scheme for the weights. Note that depending on the the basis V_Y , we get the degree of $p \in \Pi_{k-1}^d$. Below are common RBFs and their respective cpd orders:

Table 1: Common radial basis functions and their minimal CPD order.

RBF	Radial function $\phi(r)$	CPD order k
Gaussian	$\exp(-\varepsilon^2 r^2)$	0
Inverse multiquadric	$(r^2 + \varepsilon^2)^{-1/2}$	0
Multiquadric	$(r^2 + \varepsilon^2)^{1/2}$	1
Linear	r	1
Cubic	r^3	2
Thin-plate spline	$r^2 \log r$	2
Quintic	r^5	3

This order provides us both, the degree of the polynomials $k - 1$ and the number of terms:

$$l = \binom{d + k - 1}{k - 1} \quad (18)$$

As shown in [8], a polynomial augmentation can be added with a larger degree than required for the RBF, acting as a correction term while inheriting the non-oscillatory behavior of the RBF at the boundary $\partial\Omega$.

2.3 Consistency Error Analysis

We begin by defining the native space of solutions for (1) with (4). Suppose $\mathcal{F}_K = H^{s+2}(\Omega)$. This defines an RKHS for functions $u \in H^{s+2}(\Omega)$, $Lu \in H^s(\Omega)$ and by Morrey Embedding, $L(\cdot)(\mathbf{x})$ is bounded, linear functional on $H^{s+2}(\Omega)$. Thus, by Riesz, $v_{Lu,x} \in H^{s+2}(\Omega)$. We can also define a Kernel $K : \Omega \times \Omega \rightarrow \mathbb{R}$ such that

$$f(x) = \langle f, K(\cdot, x) \rangle_{\mathcal{F}_K} \quad (19)$$

$$(Lf)(x) = \langle f, L_x K(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_K} \quad (20)$$

$$\mathbf{w}^T(\mathbf{x})\mathbf{f} = \langle f, \mathbf{w}^T(\mathbf{x})\mathbf{k}(\cdot) \rangle_{\mathcal{F}_K} \quad (21)$$

$$\xi_{w,L,x}(f) := Lf(x) - \mathbf{w}^T(\mathbf{x})\mathbf{f} \quad (22)$$

$$\|\xi_{w,L,x}\|_{\mathcal{F}_K^*} := \sup_{\substack{f \in \mathcal{F}_K \\ \|f\|_{\mathcal{F}_K} \leq 1}} |\xi_{w,L,x}(f)| \quad (23)$$

L_x indicates applying the operator acted on the variable $\mathbf{x}$ in the kernel. We define $\mathbf{k}(\mathbf{z}) = [k(\mathbf{z}, \mathbf{x}_1), \dots, k(\mathbf{z}, \mathbf{x}_n)]^T$. Since $\|\xi_{w,L,x}\|_{\mathcal{F}_K^*} < \infty$, by Riesz Representation Theorem, $v_{w,L,x} = L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x})\mathbf{k}(\cdot)$ with error estimate $\|\xi_{w,L,x}\|_{\mathcal{F}_K^*} = \|v_{w,L,x}\|_{\mathcal{F}_K}$. By [5], the functional is continuous on $\mathcal{F}_K$ if $L_x K$ exists and both, polynomial consistency and operator exactness are guaranteed. Finally, We define the power function $\mathcal{P}^2(x) = \|v_{w,L,x}\|_{\mathcal{F}_K}^2$. Note that:

$$\begin{aligned}
\mathcal{P}^2(\mathbf{x}) &= \langle v_{w,L,\mathbf{x}}, v_{w,L,\mathbf{x}} \rangle_{\mathcal{F}_{\mathcal{K}}} \\
&= \langle L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot), L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_{\mathcal{K}}} \\
&= \langle L_x K(\cdot, \mathbf{x}), L_x K(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_{\mathcal{K}}} - 2 \langle L_x K(\cdot, \mathbf{x}), \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_{\mathcal{K}}} + \langle \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot), \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_{\mathcal{K}}} \\
&= L_y L_z K(y, z)|_{y=x=z} - 2 \sum_{i=1}^n w_i(\mathbf{x}) L_x K(\mathbf{x}, \mathbf{x}_i) + \sum_{i,j=1}^n w_i(\mathbf{x}) w_j(\mathbf{x}) K(\mathbf{x}_i, \mathbf{x}_j) \\
&= L_y L_z K(y, z)|_{y=x=z} - \sum_{i=1}^n w_i(\mathbf{x}) L_x K(\mathbf{x}, \mathbf{x}_i)
\end{aligned}$$

The last line is carried out by Eq (17).

3 Weight Constraint RBF-FD Setup

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to u sampled at nodes. Suppose $m > n$ (for the case of $m < n$, refer to [1]), thus more center nodes than stencil nodes per stencil. Additionally, instead of assuming exactness of the form $L\phi_j(\mathbf{x}) = \sum_i^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$ for $j = 1, \dots, m$ as in the classical RBF-FD, we define a residual function:

$$r_j(\mathbf{x}) := L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \quad (24)$$

$$r(\mathbf{x}) = L\phi(\mathbf{x}) - \Phi^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \quad (25)$$

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials augmenting the radial basis functions. Let $\Pi_k = \text{span}\{p_i\}_{i=1}^l$ with polynomial degree up to k . We want to reproduce the operator exactly on the polynomial basis:

$$Lp_j(\mathbf{x}) := \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \quad (26)$$

$$Lp(\mathbf{x}) = \mathbf{P}^T \mathbf{w} \quad (27)$$

where $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \Phi^T \mathbf{w} = Lp \end{cases} \quad (28)$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned}
\mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp(\mathbf{x}) \rangle \\
&= \frac{1}{2} \langle \Phi^T \mathbf{w}, \Phi^T \mathbf{w} \rangle - \langle \Phi^T \mathbf{w}, Lp \rangle + \frac{1}{2} \langle Lp, Lp \rangle \\
&\quad + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp(\mathbf{x}) \rangle.
\end{aligned}$$

4 Consistency Conditions for Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\nabla_w \mathcal{L} = 0 : \Phi L \phi = \Phi \Phi^T w + P \nu \quad (29)$$

$$\nabla_\nu \mathcal{L} = 0 : P^T w = L p \quad (30)$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix} \begin{bmatrix} w \\ \nu \end{bmatrix} = \begin{bmatrix} \Phi L \phi \\ L p \end{bmatrix} \quad (31)$$

From [6], Equation (31) has a unique solution given $L p \in \text{Range}(\mathbf{P}^T)$. The primary goal is to approximate $Lu(\mathbf{x}) \approx \mathbf{w}^T(\mathbf{x})\mathbf{u}$, with $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. Define $L_h f(\mathbf{x}) := \mathbf{w}^T(\mathbf{x})\mathbf{f}$ for some function f . Thus, we have the following:

$$\begin{aligned} L_h f(\mathbf{x}) &= \mathbf{w}^T(\mathbf{x})\mathbf{f} \\ &= [\mathbf{w}^T \quad \nu^T] \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} \\ &= \left(\begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} \Phi L \phi \\ L p \end{bmatrix} \right)^T \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} \\ &= [L \phi^T \Phi^T \quad L p^T] \begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix}^{-T} \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} \\ &= [L \phi^T \Phi^T \quad L p^T] \begin{bmatrix} \mathbf{q} \\ \alpha \end{bmatrix} \\ &= L \phi^T(\mathbf{x}) (\Phi^T \mathbf{q}) + L p^T(\mathbf{x}) \alpha \end{aligned}$$

Next, we can show that the coefficients $\mathbf{q}, \alpha$ relate to the coefficients of u as follows:

$$\begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \alpha \end{bmatrix} \quad (32)$$

Define $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\Phi^T)\} \subseteq V_Y$. Let $s \in V_h + \Pi_k$. Thus, $s(x) = \phi^T(\mathbf{x})\Phi^T \mathbf{q} + p^T(\mathbf{x})\alpha$ where $\mathbf{q} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}^l$. Then:

$$\mathbf{s} = s(X) = \Phi \Phi^T \mathbf{q} + P \alpha = f(X) = \mathbf{f} \quad (33)$$

$$P^T \mathbf{q} = 0 \quad (34)$$

Hence, we get interpolation of f at the node set X . Furthermore, from the KKT matrix we can show that $\Phi \mathbf{r} = P \nu$ which implies that $\Phi \mathbf{r} \in \text{range}(P) = (\ker(P^T))^\perp$. Thus, we get the relationship:

$$\mathbf{r}^T (\Phi^T \mathbf{z}) = 0 \text{ for } P^T \mathbf{z} = 0 \quad (35)$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\mathbf{P}^T \mathbf{q} = \mathbf{0}$.

$$\begin{aligned}
Ls(x) - L_h s(x) &= L(\phi^T(x) \Phi^T \mathbf{q} + \mathbf{p}^T(x) \alpha) - \mathbf{w}^T(x) s \\
&= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \alpha - \mathbf{w}^T(x) (\Phi \Phi^T \mathbf{q} + \mathbf{P} \alpha) \\
&= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \alpha - \mathbf{w}^T(x) \Phi \Phi^T \mathbf{q} - \mathbf{w}^T(x) \mathbf{P} \alpha \\
&= (L\phi(x) - \Phi^T \mathbf{w}(x))^T \Phi^T \mathbf{q} + (L\mathbf{p}^T(x) - \mathbf{P}^T \mathbf{w}(x))^T \alpha \\
&= \mathbf{r}^T \Phi^T \mathbf{q} + (L\mathbf{p}(x) - \mathbf{P}^T \mathbf{w}(x))^T \alpha
\end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (30) and (35)} \quad (36)$$

5 Consistency Results for Function Approximations

Let u be a solution to (1) and $s \in V_h + \Pi_k$. We can say $u = s + (u - s)$ to get the following:

$$\begin{aligned}
\mathcal{E}[u](\mathbf{x}) &:= Lu(\mathbf{x}) - L_h u(\mathbf{x}) \\
&= Lu(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} \\
&= L((u - s) + s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} - L_h s(\mathbf{x}) + L_h s(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) + (Ls(\mathbf{x}) - L_h s(\mathbf{x})) + L_h(s - u)(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) (\mathbf{u} - \mathbf{s}), \text{ by (36)} \\
&= Le(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{e}, \mathbf{e}(\mathbf{x}) = u(\mathbf{x}) - s(\mathbf{x}) \\
&= \mathcal{E}[\mathbf{e}](\mathbf{x})
\end{aligned}$$

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